

HARSH UPADHYAY

Bengaluru, Karnataka +91-8006956598 harshupadhyay1414@gmail.com [LinkedIn](#) [GitHub](#)

SUMMARY

B.Tech CSE graduate with hands-on experience building AI/ML systems and LLM-powered applications. Skilled in end-to-end model development — from data preprocessing and feature engineering to production deployment. Seeking a data science or ML engineering role to deliver measurable impact.

EDUCATION

Hindustan College of Science and Technology, Mathura 2022 – 2026
B.Tech in Computer Science and Engineering with **Minor in Project Management** **CGPA: 6.44**
Relevant Coursework: Data Structures & Algorithms, OOP, Database Management Systems, Machine Learning

WORK EXPERIENCE

Aimerz.ai (via Mentormind Program) Aug – Sep 2025
Machine Learning Intern Remote

- Built Uber fare regression model (XGBoost + Random Forest) on 500K+ records; achieved ~90% R^2 with feature engineering (time-of-day, surge zone, distance bins); reduced RMSE by 18% over baseline.
- Performed EDA, outlier removal, and feature transformation using Pandas and Scikit-learn.

India Space Lab Jun – Jul 2025
AI and Machine Learning Intern Remote

- Achieved 85%+ accuracy on anomaly classification using Random Forest and SVM on real-world datasets.
- Automated preprocessing pipeline (imputation, normalization, train-test split) reducing manual prep time by 60%.

PROJECTS

Breast Cancer Detection System (CAD) — Capstone Project [GitHub](#)
Tech Stack: Python, TensorFlow/Keras, Scikit-learn, Flask, DDSM Dataset

- Dual-pipeline CAD: deep learning (DenseNet121, **EfficientNetB0**, ResNet50, MobileNetV2) with two-phase transfer learning and 3-fold patient-level cross-validation to prevent data leakage.
- Classical ML (GLCM/shape features + SMA-AOA optimizer + ensemble) achieved **94.2% AUC**; Grad-CAM heatmaps via Flask for radiologist-interpretable predictions.

GenAI Resume–JD Matching & Skill Gap Analyzer [Live Demo](#) | [GitHub](#)
Tech Stack: Python, FastAPI, FAISS, Sentence Transformers, Groq LLaMA 3.3 70B, PyMuPDF

- Unlike keyword-based ATS tools, used **FAISS + Sentence Transformers** for meaning-based skill matching across 9 categories (sub-second); 4-tool career suite (cover letter, ATS rewriter, skill gap, interview prep) via **5 modular FastAPI endpoints** with graceful LLM fallback.

AI Financial News Analyzer (LLM + RAG) [Live Demo](#) | [GitHub](#)
Tech Stack: Python, FastAPI, Flask, Groq LLaMA 3, FAISS, Plotly, NLP

- Finance-specific NLP pipeline: ticker extraction (20+ symbols/query), 3-class sentiment classification (75% accuracy), and Plotly market charts; **RAG layer** via FAISS grounds answers with source attribution, eliminating hallucinated financial data; sub-500ms via FastAPI.

DocMind — PDF Q&A Bot & Article Summarizer [Live Demo](#) | [GitHub](#)
Tech Stack: Python, Flask, Groq LLaMA 3, PyMuPDF, BeautifulSoup, ReportLab

- Multi-source context fusion** — ingests multiple PDFs and live web articles (BeautifulSoup) into a unified context window for cross-document Q&A; session history and PDF export via ReportLab; sub-3s on Render.

SKILLS

Languages	Python (DSA, OOP, NumPy, Pandas), SQL
Machine Learning	Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Scikit-learn, XGBoost
Deep Learning	CNN, ANN, Transfer Learning, Grad-CAM, TensorFlow, Keras
Generative AI & NLP	LLMs, Prompt Engineering, RAG, FAISS, Sentence Transformers, Groq API
Data Analytics	EDA, Power BI (DAX, Power Query), MS Excel (Pivot Tables, XLOOKUP)
Databases	MySQL, PostgreSQL, MongoDB Atlas, SQLite
Cloud & Deployment	AWS (EC2, S3), Render, Flask, FastAPI, Streamlit
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code